

Government College of Engineering, Jalgaon
(An Autonomous Institute of Government of Maharashtra)

Name :	PRN :
Class : L. Y. B.Tech Computer	Batch:
Subject : CO456U Cloud Computing Lab	Sem : VIII th
Course Teacher: Prof. S. D. Cheke	Academic Year : 2025-26
Date of Performance :	Date of Completion :

Practical No. 8

Aim: Create POST, GET end point using AWS Api Gateway and integrate that with lambda function.

Theory :

Objective: To create REST API endpoints (GET and POST) using API Gateway and integrate them with an AWS Lambda function to process requests.

What is AWS Lambda?

AWS Lambda is a serverless computing service that allows you to run code without managing servers.

- Executes code on demand
- Automatically scales
- Charges only for execution time

What is API Gateway?

Amazon API Gateway is a service that allows developers to create, publish, and manage APIs.

- Acts as an entry point for client requests
- Supports REST and HTTP APIs
- Connects frontend applications with backend services like Lambda

What are GET and POST Methods?

- **GET** → Used to retrieve data
- **POST** → Used to send/store data

Integration Flow

Client → API Gateway → Lambda → DynamoDB → Response

- Client sends request
- API Gateway triggers Lambda
- Lambda processes request and interacts with database
- Response sent back to client

What is DynamoDB?

Amazon DynamoDB is a fully managed NoSQL database service by AWS.

- Stores data in key-value format

- Highly scalable and fast
- No server management required

Requirements

- AWS account
- Existing Lambda function (already created ✓)
- DynamoDB table

Procedure / Steps

Step 1: Open API Gateway

1. Go to AWS Console
2. Open API Gateway
3. Click **Create API**

Step 2: Choose API Type

- Select **HTTP API** (simpler)
- Click **Build**

Choose an API type [Info](#)

HTTP API

Build low-latency and cost-effective REST APIs with built-in features such as OIDC and OAuth2, and native CORS support.

Works with the following:
Lambda, HTTP backends

[Import](#)

[Build](#)

Step 3: Add POST Route

1. Click **Add Integration**
2. Choose:
 1. Integration type: Lambda
 2. Select your function: insertUser
3. Click **Add Route**
 - Method: POST
 - Resource path: /user

Configure routes - *optional*

Configure routes [Info](#)

API Gateway uses routes to expose integrations to consumers of your API. Routes for HTTP APIs consist of two parts: an HTTP method and a resource path (e.g., GET /pets). You can define specific HTTP methods for your integration (GET, POST, PUT, PATCH, HEAD, OPTIONS, and DELETE) or use the ANY method to match all methods that you haven't defined on a given resource.

Method	Resource path	Integration target	
POST ▼	/user	insertUser ▼	Remove
GET ▼	/user	insertUser ▼	Remove

[Add route](#)

[Cancel](#)

[Review and create](#)

[Previous](#)

[Next](#)

Configure API

API details

API name

An HTTP API must have a name. The name is a non-unique value you use to identify and organize your APIs. To programmatically refer to this API, use the API ID that API Gateway generates for you.

IP address type [Info](#)

Select the type of IP addresses that can invoke the default endpoint for your API. You don't need to redeploy your API for the update to take effect.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

Integrations (1) [Info](#)

Specify the backend services that your API will communicate with. These are called integrations. For a Lambda integration, API Gateway invokes the Lambda function and responds with the response from the function. For an HTTP integration, API Gateway sends the request to the URL that you specify and returns the response from the URL.

AWS Region

Lambda function

Version

[Learn more](#)[Add integration](#)[Cancel](#)[Review and create](#)[Next](#)

Step 4: Add GET Route

1. Click **Add Route**
2. Configure:
 1. Method: GET
 2. Resource path: /user
3. Attach same Lambda function

APIs (1/1)

	Name	Description	ID	Protocol	API endpoint type	Created
☐	rjtest		1pbhvl3mw4	HTTP	Regional	2026-04-15

Stage details

[Delete](#)[Edit](#)

Details

Name

\$default

Created

April 16, 2026 12:36 AM

Last updated

April 16, 2026 12:36 AM

Invoke URL

<https://1pbhvl3mw4.execute-api.eu-north-1.amazonaws.com>

Update Lambda for GET & POST

```
import json
```

```
import boto3
```

```
dynamodb = boto3.resource('dynamodb')
```

```
table = dynamodb.Table('Users')
```

```

def lambda_handler(event, context):

    method = event['requestContext']['http']['method']

    try:

        if method == "POST":

            body = json.loads(event['body'])

            user_id = body['user_id']

            name = body['name']

            table.put_item(

                Item={

                    'user_id': user_id,

                    'name': name

                }

            )

            return {

                'statusCode': 200,

                'body': json.dumps('User added successfully')

            }

        elif method == "GET":

            user_id = event['queryStringParameters']['user_id']

            response = table.get_item(Key={'user_id': user_id})

            return {

```

```
'statusCode': 200,  
  
    'body': json.dumps(response.get('Item', {}))  
  
}
```

except Exception as e:

```
    return {  
  
        'statusCode': 500,  
  
        'body': str(e)  
  
    }
```

Testing

POST Request (Insert Data)

Use Postman:

- Method: POST
- URL: <https://1pbhvl3mw4.execute-api.eu-north-1.amazonaws.com/user>

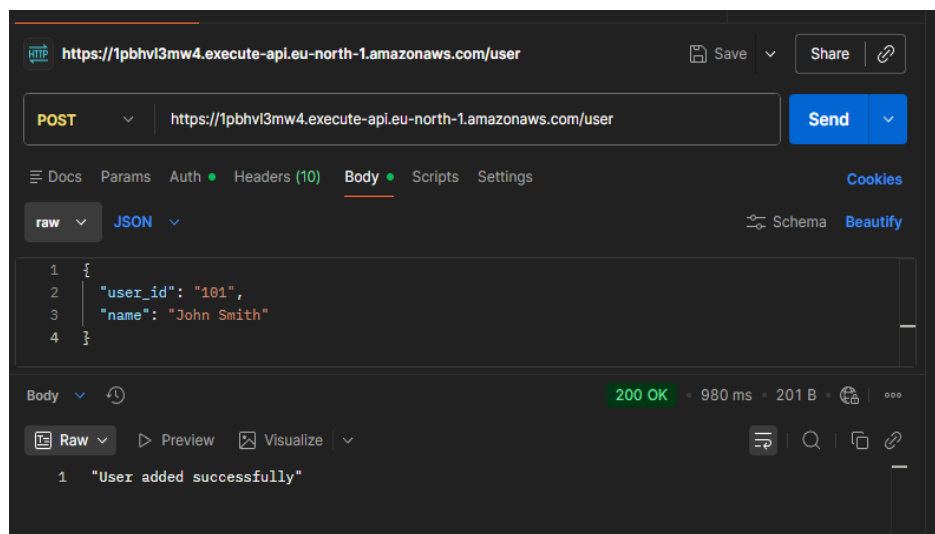


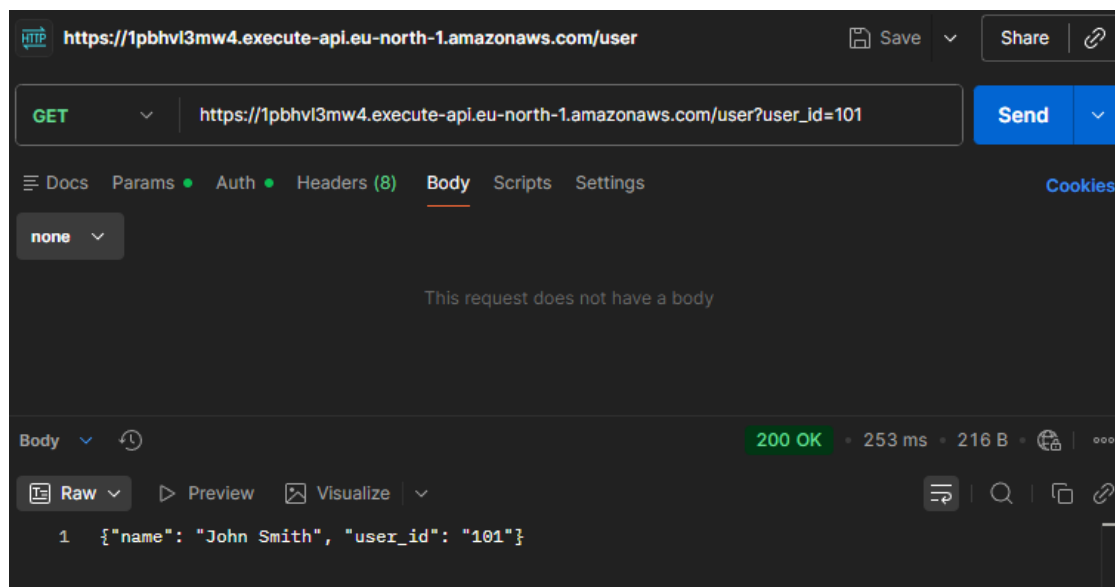
Table: users - Items returned (1)

Scan started on April 16, 2026, 00:48:55

☐	user_id (String)	▼	name
☐	101		John Smith

GET Request (Fetch Data)

- Method: GET
- URL: https://1pbhvl3mw4.execute-api.eu-north-1.amazonaws.com/user?user_id=101



Successfully created GET and POST endpoints using API Gateway and integrated them with Lambda to store and retrieve data from DynamoDB.

Conclusion: In this way, we have successfully implemented and completed the creation of GET and POST endpoints integrated with a Lambda function.

Prof. S. D. Cheke
Course Teacher